

Clustering for Big Data Analytics

Group Members

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- Group 4 – Clustering
- Dataset: CIC-IDS2017
(Monday & Friday focus)



Why Clustering Is Needed in Big Data

- Big data often lacks labels or has inconsistent labeling
- High-dimensional feature spaces (80+ features in CIC-IDS2017)
- Extreme class imbalance – supervised models fail
- Discovery of unknown attack patterns



Overview of Clustering

- Unsupervised learning: groups data by similarity
- No labeled data required
- Ideal for large, high-dimensional datasets



CIC- IDS2017 Dataset Summary

- Created by Canadian Institute for Cybersecurity (UNB)
- 5 days of network traffic: 2.8M total records
- 80+ flow features extracted using CICFlowMeter
- Contains both benign and multiple attack types

Dataset Focus: Monday Traffic

- Primary attacks: FTP-Patator, SSH-Patator
- 83% benign traffic dominates dataset
- Brute force attacks are low volume (1-2%)
- Sequential/temporal patterns important for detection



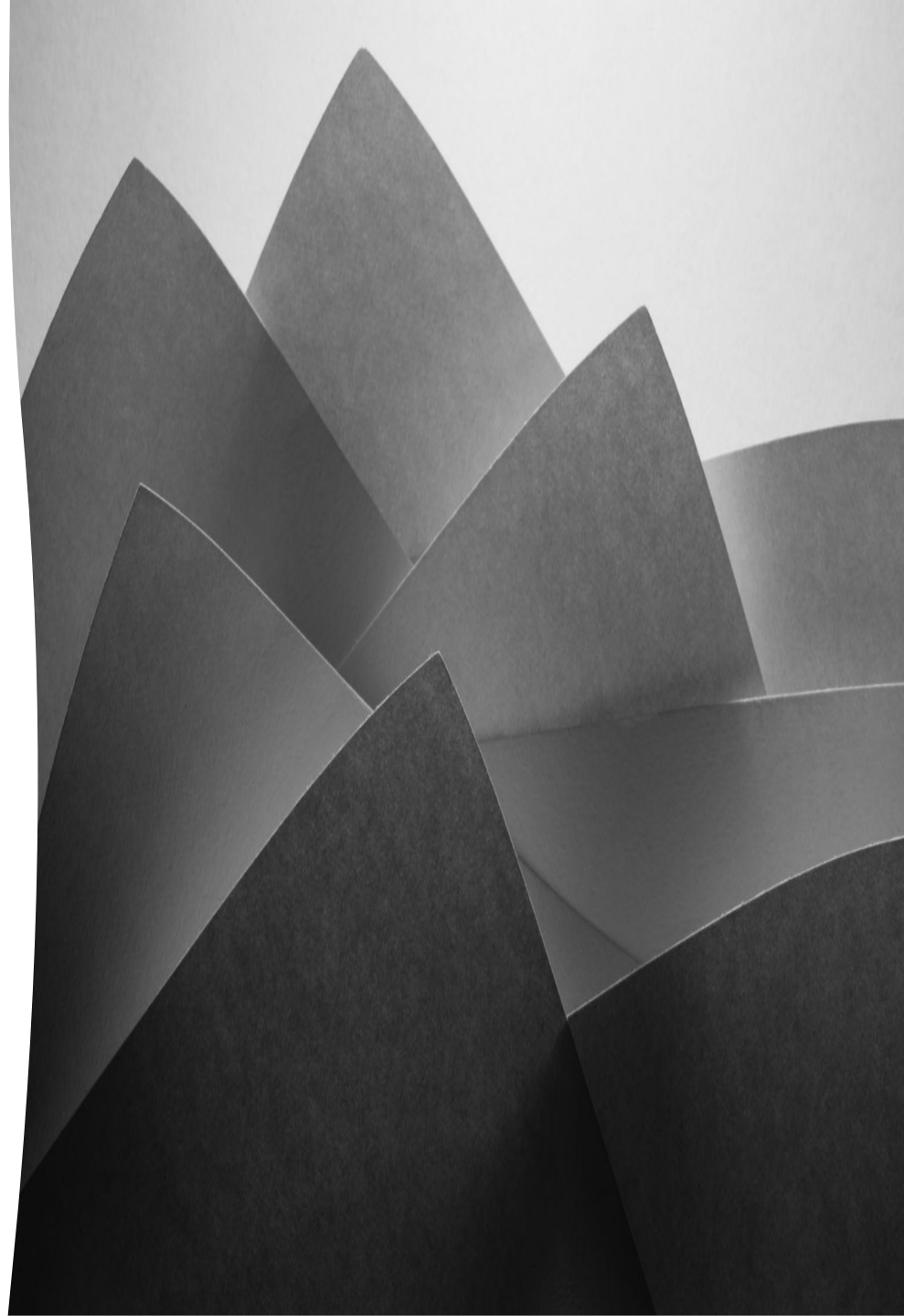
Dataset Focus: Friday Traffic

Attacks: DDoS,
PortScan, Botnet

Botnet
extremely rare
(~0.87%)

Reverse
imbalance: DDoS
& PortScan >
benign

Multiple
simultaneous
attacks increase
complexity



Feature Space Characteristics

- 80+ numerical features (flow duration, IATs, flags, packet stats)
- Temporal features capture attack patterns
- High dimensionality → clustering separates behavior profiles
- Different attacks leave distinct statistical signatures



Why Supervised Learning Fails Here

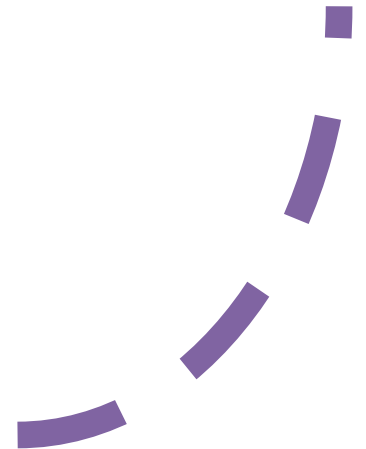
- Severe majority class dominance (e.g., 75:1 ratio)
- Models predict 'benign' for most samples → misleading accuracy
- Rare attacks (Botnet, Heartbleed, Infiltration) ignored
- Requires heavy resampling → introduces noise





Why Clustering Works Well

- Labels not required
- Naturally handles class imbalance
- Identifies anomalies & rare attack behavior
- Detects new or evolving attack patterns (zero-days)



K-Means for Monday Traffic

- Suitable when clusters are well-separated
- Expected groups: benign, FTP-Patator, SSH-Patator
- Brute-force attacks form high-frequency clusters
- Efficient for >500K Monday records

```
=====
STEP 4: K-MEANS CLUSTERING
=====
```

```
Training K-Means with k=4...
Please wait, this may take 2-3 minutes...
✓ Clustering complete!
```

```
=====
SILHOUETTE SCORE: 0.5747
=====
```

```
Interpretation:
```

```
> 0.70: Excellent clustering
> 0.50: Good clustering
> 0.25: Fair clustering
< 0.25: Poor clustering
```

DBSCAN for Friday Traffic

- Density-based: identifies clusters & outliers
- Botnet traffic appears as sparse anomaly points
- Effective for irregular cluster shapes (DDoS, PortScan)
- Does not require pre-selecting number of clusters

Clustering Workflow for CIC- IDS2017

- 1. Data cleaning & normalization
- 2. Feature selection / PCA dimensionality reduction
- 3. Apply K-Means, DBSCAN, Isolation Forest
- 4. Analyze clusters → identify attack signatures
- 5. Validate clusters using labels (post-clustering check)

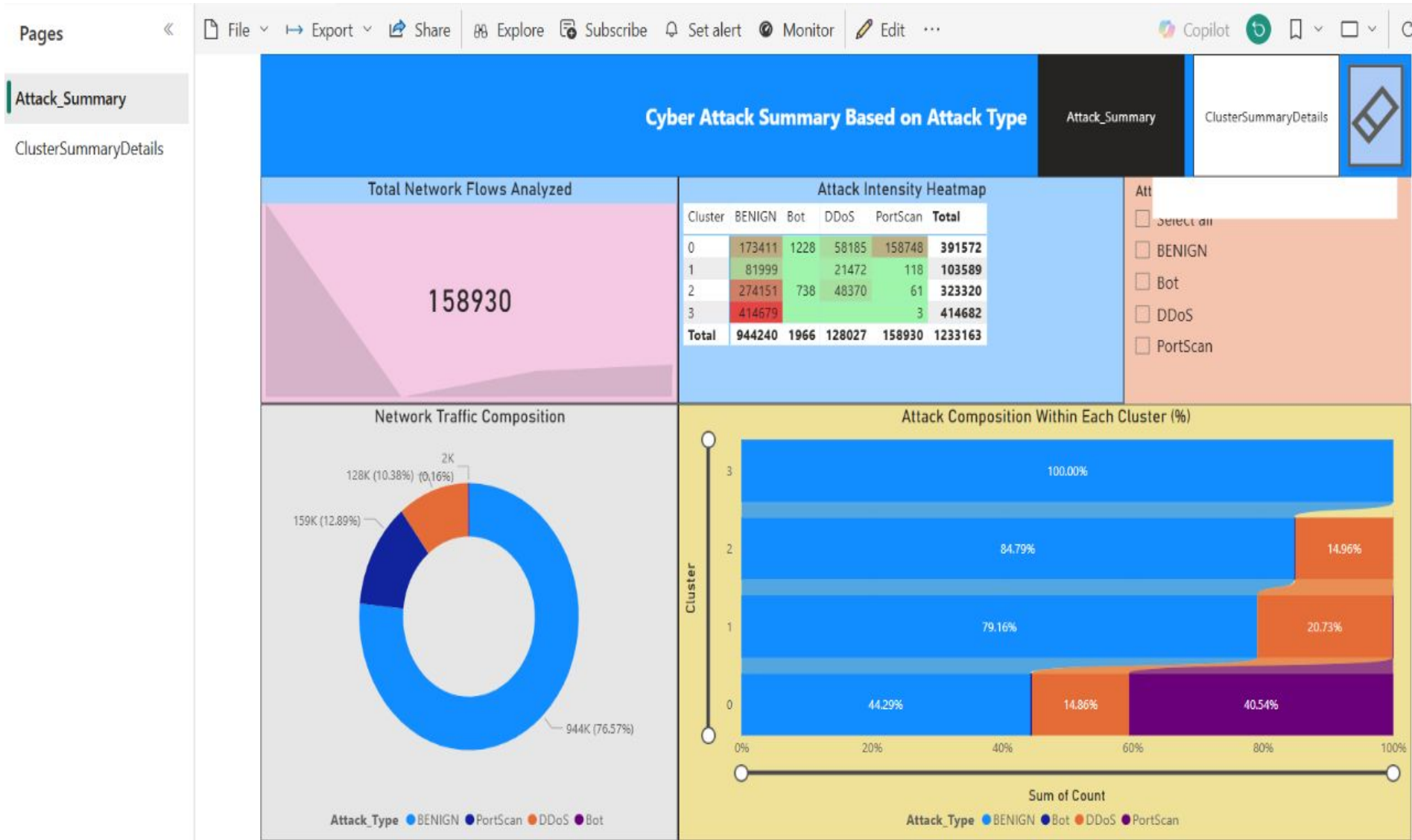
Clustering Analysis

CLUSTER ANALYSIS

--- CLUSTER SIZE DISTRIBUTION ---

+-----+-----+	
cluster	count
+-----+-----+	
0	391572
1	103589
2	323320
3	414682
+-----+-----+	

Dashboards



Attack_Summary

ClusterSummaryDetails

Attack_Type

☐ BENIGN

☐ Bot

☐ DDoS

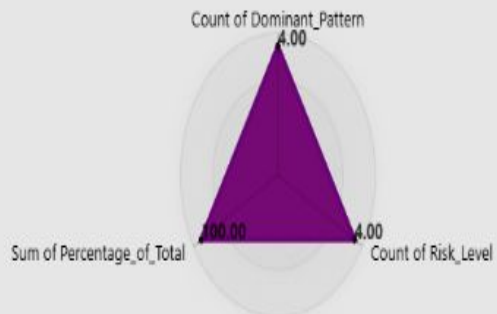
Security Risk level & attack distribution

Attack_Summary

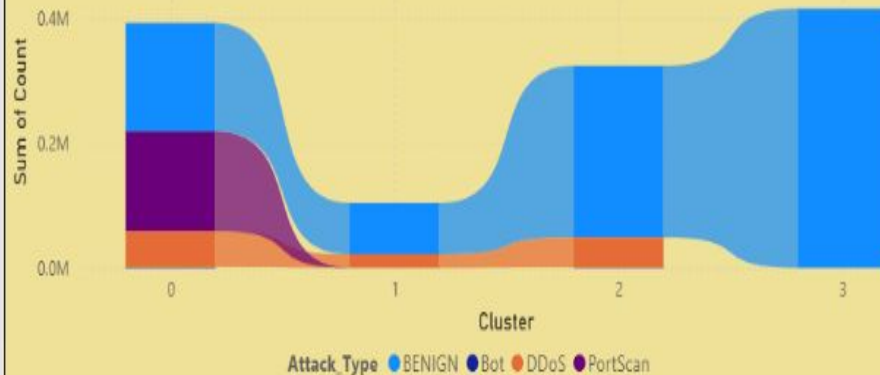
ClusterSummaryDetails



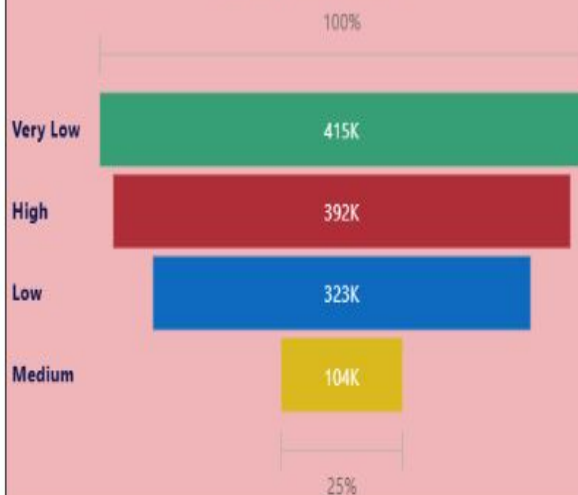
Attack_Type ● BENIGN ● Bot ● DDoS ● PortScan



Attack Distribution Flow Across Clusters"



Security Risk Funnel



Interactive Attack Analysis Explore



Attack_Type

☐ BENIGN

☐ Bot

☒ DDoS

Security Risk level & attack distribution

Attack_Summary

ClusterSummaryDetails

Attack_Type ● DDoS

Count of Dominant_Pattern

4.00

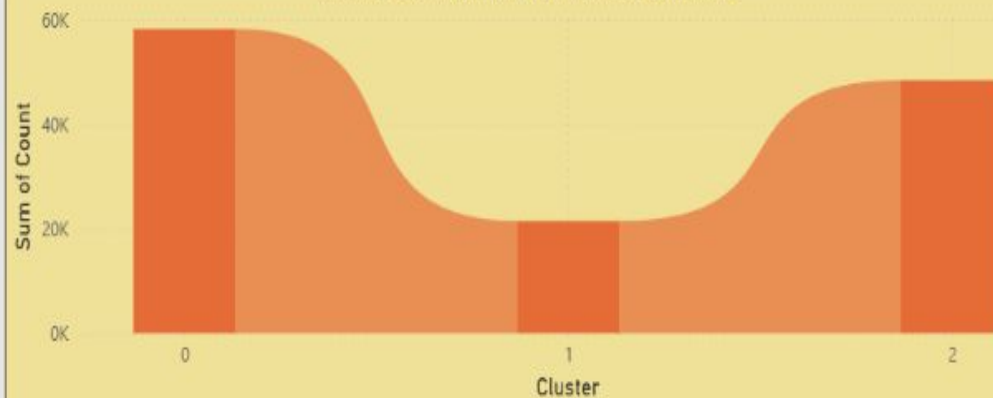
Sum of Percentage_of_Total

100.00

Count of Risk_Level

4.00

Attack Distribution Flow Across Clusters"



Attack_Type ● DDoS

Security Risk Funnel

100%

Very Low

415K

High

392K

Low

323K

Medium

104K

25%

Interactive Attack Analysis Explore

Attack_Type x

DDoS

Cluster x

1

Percentage x

Sum of Count

128027

DDoS

128027

0

58185

2

48370

1

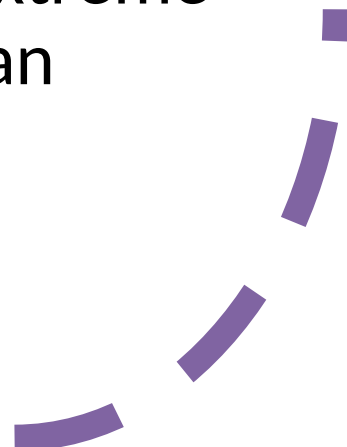
21472

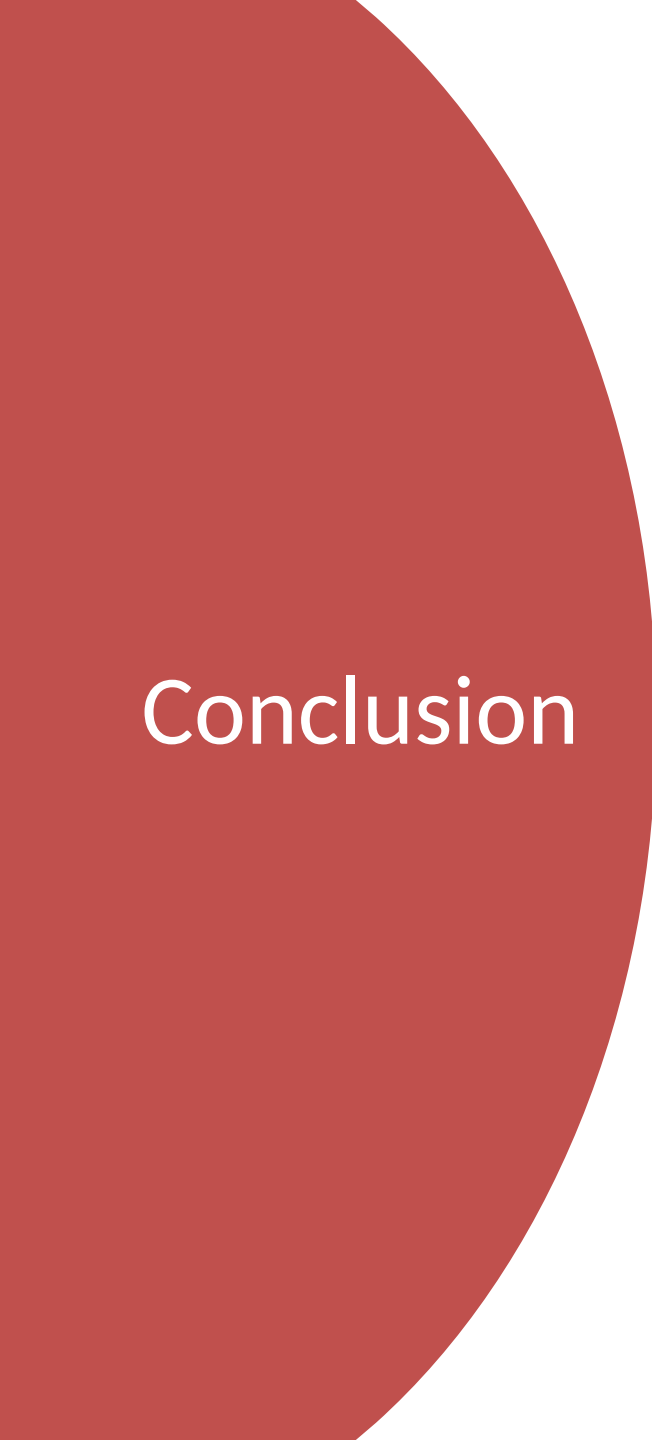
20.73

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Key Findings

- Monday brute-force attacks form distinct statistical clusters
 - High-dimensional features naturally separate attack categories
 - Clustering handles extreme imbalance better than classifiers
- 



Conclusion

- Clustering is essential for modern big data environments
 - It discovers unknown patterns without labels
 - Ideal for cybersecurity, fraud detection, segmentation
 - Supports scalable, adaptive analytics for evolving datasets
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